

Anne Kaitlyn Arabelo

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EDUCATION

University of California, San Diego | B.S. in Bioengineering *Expected Fall 2027*
Coursework: Bioengineering Laboratory, Bioengineering Physiology I & II, Probability & Statistics, Biomaterials Design

EXPERIENCE

Research Assistant | Mittal Lab (PI: Ravinder Mittal, M.D.), UC San Diego *Jul 2025 – Present*

- Developed a Python tool that processes patient esophageal motility data and automatically generates matplotlib visualizations of disease progression in eosinophilic esophagitis (EoE)
- Produced figures and analysis for a co-authored abstract presented at Digestive Disease Week (DDW) 2026
- Audited IRB-approved clinical study datasets, identifying and correcting inconsistencies to maintain data integrity
- Maintained participant records and study documentation in compliance with IRB protocols and HIPAA requirements

Research Assistant | Aran Lab (PI: Kiana Aran, Ph.D.), UC San Diego *Sep 2024 – Jun 2025*

- Designed microfluidic blood-brain barrier (BBB) chips in SolidWorks and laser-cut them for iPSC neural cultures
- Integrated electrodes into the BBB chip to measure barrier integrity via transendothelial electrical resistance (TEER)
- Collaborated with lab members on iPSC-derived cell culture to test chip prototypes
- Wrote a literature review on organ-on-chip BBB models and presented findings in a research poster

PROJECTS

Biomechanical HRIM: Quantifying Esophageal Mechanical Work | Senior Design, Mittal Lab *Sep 2025 – Jun 2026*

- Built, with a 5-person team, a MATLAB pipeline that converts high-resolution impedance manometry (HRIM) pressure and impedance data into biomechanical swallow metrics (cross-sectional area, tension, stiffness, mechanical work)
- Contributed to data analysis of pre- and post-dupilumab HRIM studies from 10 eosinophilic esophagitis (EoE) patients, showing significantly reduced mechanical work in 5 of 6 bolus conditions, most pronounced in the distal esophagus
- Validated impedance-derived cross-sectional area against ultrasound, with strong agreement before and after treatment
- Evaluated a reformulated conductive gel for HRIM testing, tracking conductivity and viscosity (rotational viscometer) over 45 days to improve shelf life and measurement consistency

PRESENTATIONS

Co-author, EoE research abstract presented at Digestive Disease Week (DDW) 2026, Chicago, IL *May 2026*
Team poster on biomechanical HRIM analysis in EoE, UC San Diego Bioengineering Day, La Jolla, CA *2026*

SKILLS

Laboratory: Microfluidic device design and fabrication (SolidWorks, Glowforge laser cutting), electrode integration, TEER measurement, size-exclusion chromatography, centrifugation, hematocrit

Computational: Python (matplotlib), MATLAB, Git/GitHub, statistical analysis

Clinical Research: Medtronic ManoView (high-resolution manometry), ultrasound data, participant records

Training: CITI Human Subjects Research, HIPAA, Laboratory Safety

LEADERSHIP & ACTIVITIES

Sub-Lead | Biomedical Engineering Society (BMES), UC San Diego *Sep 2024 – Present*

- Co-organized networking events linking 50+ students with 10 companies, including Philips, Medtronic, and BD
- Recruited and coordinated 7 industry professionals to participate in chapter networking events
- Collaborated with a 7-person team in weekly meetings to plan events, divide responsibilities, and track timelines